

Alexander Braverman

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Education

Cornell University

Ithaca, NY

B.S. in Computer Science and Operations Research (Double Major) | GPA: 3.88/4.00

Expected May 2028

- **Coursework:** Reinforcement Learning, Machine Learning, Robot Learning, Probability & Statistics, Data Structures

Research Experience

UCLA Artificial General Intelligence Lab

Los Angeles, CA

Natural Language Processing Researcher, First Author

Dec. 2024 – Jul. 2026

- Built **Test-Time A* Search (TTA*)**, a training-free decoding wrapper that reframes multi-step reasoning as depth-regularized tree search over complete candidate solutions; raised accuracy on 5 open SLMs (1–8B, Llama + Qwen) across 4 math benchmarks by **+10.6–13.8 pts on GSM8K** and **+9.6–27.4 pts on MATH500** over zero-shot CoT
- Outperformed every configuration of Best-of-*N*, Iterative Self-Refine, and Tree-of-Thoughts at any compute budget swept: **95.4% vs. 92.0%** best baseline on GSM8K (Qwen3-4B) using **4× fewer generated tokens** (6.1k vs. 23.4k per problem)
- Served Qwen3-4B behind a vLLM OpenAI-compatible endpoint with continuous batching on NVIDIA RTX A6000s via Slurm array jobs; batched independent self-evaluation samples within each node expansion to recover throughput lost to TTA*'s inherently sequential search
- Built a benchmarking harness sweeping **4 methods × 6 budgets × 5 models** over 2,250 problems, instrumenting per-problem model calls, generated tokens, and wall-clock to produce accuracy-vs-compute Pareto curves
- Full paper submitted to EACL 2027; code open-sourced at github.com/AlexBraverman/test-time-a-star; invited to present at Y Combinator ML Research Symposium (2026)

UNC Chapel Hill Zhang Lab

Remote

Reinforcement Learning Researcher, First Author

Dec. 2025 – Present

- Showed that **causal influence exploration** fails in POMDPs where the true state is unobserved; developing a causal exploration algorithm that tells the agent which objects it can actually influence without observing true state, with guarantees for when that judgment is reliable
- Building an exploration bonus that matches the original under clean sensors and stays reliable when they degrade: a PyTorch ensemble of transition models over a GRU belief encoder, with experiments run on MiniGrid and MuJoCo manipulation tasks
- Paper to be submitted to ICML 2027; advised by Prof. Weitong Zhang

UCLA Artificial General Intelligence Lab

Los Angeles, CA

Natural Language Processing Researcher, First Author

Apr. 2023 – Dec. 2024

- Built **explanatory prompting**, a method that hands the LLM an informal algorithmic description (DFS) of a task class; cut hallucination rate on a DAG-connectivity benchmark from **44.8% to 1.8%** (20.8% → 98.2% accuracy, GPT-4)
- Engineered a Python/OpenAI-API evaluation harness running 5 prompting arms × 500 instances plus 2,000 sampled generations, with automated answer parsing and per-arm accuracy logging
- Ablated each prompt component to isolate its contribution: topological-order hint **70.2%**, full algorithmic intuition **95.2%**, one-shot worked example **98.2%**, vs. **55.2%** for the zero-shot CoT baseline

University of Maryland FROOT Lab

College Park, MD

Computer Networking Researcher, First Author

Apr. 2022 – Sep. 2023

- Built a Bloom-filter-based migration telemetry layer in C++ for live Redis shard migration, cutting query loss rate **3.5–3.8×** (11.8% → 3.7%) across 750K-query benchmarks on a **100GB / 50M key-value** store
- Re-implemented DistCache's migration path as a UDP-based transfer protocol streaming 50M 2KB records between source and destination clusters using 10 concurrent sender/receiver threads, with the destination sharded across 5 Redis instances
- Replaced the standard double-query pattern with an **O(1)** destination-side membership probe (integer hash, 125MB bitmap) that forwards unmigrated keys to the source in a single hop, **halving** per-request network cost
- Benchmarked on a 3-node testbed (2× Intel Xeon Gold 5317, 512GB DRAM, 10G NIC) across Zipf skews 0.90/0.95/0.99 and write ratios 0.01–0.05; measured filter error at **<0.043%** of all queries

Publications

- A. Braverman, W. Zhang, Q. Gu. “**Test-Time Scaling for Multistep Reasoning in Small Language Models via A* Search.**” *NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models.*
- A. Braverman, W. Zhang, Q. Gu. “**Mitigating Hallucination in Large Language Models with Explanatory Prompting.**” *NeurIPS 2024 Workshop on Statistical Foundations of LLMs and Foundation Models.*
- A. Braverman, Z. Liu. “**Toward Fast Query Serving in Key-Value Store Migration with Approximate Telemetry.**” *ACM SIGMETRICS Performance Evaluation Review*, 2023.

Projects

remapd github.com/dlin14/remapd | Python, LangGraph, Claude API, FastAPI, Next.js

Apr. 2026

- **Finalist, ~50 teams:** built a redistricting engine (FastAPI + Next.js) that scores congressional maps for all 50 states on population balance, compactness, and voting-rights proxies
- Wrote a simulated-annealing optimizer over county-to-district assignments in NumPy: **+27% median** objective vs. random baselines (18–42% across 5 states), **under 1s** for 700 iterations on 254 counties
- Owned the agent layer: 5-node LangGraph pipeline turning metric vectors into Claude-generated memos, streamed to the browser, with every number traceable to a verifiable US Census API call